

SENIOR FEA ENGINEER — NONLINEAR FEA & STRUCTURAL ANALYSIS

THANG NGUYEN HUU

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PROFESSIONAL SUMMARY

Structural Analysis Engineer specializing in nonlinear finite element analysis (FEA) and structural performance evaluation. Experienced in advanced modeling of contact behavior, material nonlinearity, and stability analysis with strong correlation to experimental testing. Published author with a solid computational mechanics foundation and industry experience supporting product qualification and design optimization.

TECHNICAL SKILLS

Finite Element Analysis

Abaqus, ANSYS — linear/nonlinear analysis, contact, material nonlinearity, stability, heat transfer, validation against testing.

Programming

C# (.NET, WPF, MVVM), Python (Abaqus scripting), C++ (Qt, API integration), VBA.

Engineering Tools

AutoCAD, NX, SolidWorks, Inventor; CAD automation; technical documentation with LaTeX.

PROFESSIONAL EXPERIENCE

Senior FEA Engineer

01/2025 — Present

Simpson Strong-Tie, Ho Chi Minh City

- Lead nonlinear finite element simulations for structural connectors and anchoring systems, ensuring strong correlation with full-scale physical testing.

- Review and validate complex FEA models to ensure technical accuracy and compliance with internal and external standards.
- Coordinate simulation planning and task allocation across cross-functional teams to meet project deadlines.
- Develop internal automation tools and engineering applications to improve modeling efficiency and workflow reliability.
- Provide technical mentorship and training on advanced FEA methodologies and best practices.

FEA Engineer

05/2021 — 12/2024

Simpson Strong-Tie, Ho Chi Minh City

- Performed nonlinear FEA simulations in Abaqus for structural connectors and anchoring systems under static and stability loading conditions.
- Developed Python-based Abaqus plug-ins to automate model generation and post-processing.
- Correlated numerical simulations with experimental data to validate structural performance.
- Prepared technical reports and supported internal knowledge-sharing initiatives.

SELECTED PROJECTS

Heat Transfer Parameter Identification Using Inverse Analysis

- Developed a custom 2D heat transfer finite element solver in MATLAB.
- Implemented Grey Wolf Optimization for inverse parameter identification.
- Applied POD-RBF model reduction to significantly reduce computational cost.

MATLAB

FEM

Optimization

POD-RBF

FEA Viewer (C++ / Qt)

- MVC-based desktop app for reading and visualizing Abaqus *.odb files.
- C++ data conversion kernel using the Abaqus API.
- VTK integration for 3D interactive visualization and post-processing.

C++

Qt

Abaqus API

VTK

WinUX (C# WPF)

- MVVM desktop app for secure file transfer (SFTP) between Windows and Linux servers.
- Remote job monitoring and execution control for simulation workflows.
- Real-time communication features to improve server interaction efficiency.

Engineering Automation

- Python Abaqus plug-ins for automated model generation and post-processing.
- CAD automation and workflow tools improving modeling reliability.

Python

Abaqus Scripting

CAD Automation

C#

WPF

MVVM

SFTP

PUBLICATIONS

Nguyen, M. N., **Nguyen, T. H.**, & Bui, T. Q. (2023).

Finite element modeling of tapered concrete-filled steel tubular columns under axial compression.

VNUHCM Journal of Engineering and Technology, 4(4).

<https://doi.org/10.32508/stdjet.v4i4.937>

EDUCATION

Bachelor of Engineering in Engineering Mechanics

2021

Ho Chi Minh City University of Technology, Vietnam

Relevant Coursework: Finite Element Method, Structural Analysis, Numerical Methods

LANGUAGES

English — Professional Working Proficiency

Vietnamese — Native